Exposé: Navigating the Privacy-Robustness-Performance-Trilemma in Federated Learning

An Empirical Study of Trade-offs and Combination Effects

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June 6, 2026

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1 Motivation

Federated Learning (FL) is a paradigm that allows for training machine learning models across distributed data sources without centralizing training data — a property of particular value in privacy-sensitive domains such as healthcare, finance, and mobile applications. However, the decentralized nature of FL introduces a fundamental structural tension between three system-level properties (Table 1).

Property	Provides	Cost
Privacy	Limited information leakage, protection of client data	Accuracy loss, slower convergence
Robustness	Decreased vulnerability to Byzantine or poisoning attacks	Gradient distortion → Accuracy loss
Performance	Increased model utility and efficiency	drawbacks in privacy or robustness

Table 1: Properties of FL Frameworks

Allouah et al. [1] provided the first tight theoretical lower bound demonstrating that no algorithm can simultaneously achieve strong differential privacy (DP), Byzantine robustness, and low utility loss — a result formalized as the Privacy–Robustness–Utility Trilemma. Complementing this, a 2023 position paper [2] explicitly identifies the gap between theoretical guarantees and empirical scalability as one of the most urgent open problems in the field, calling for joint benchmarks that expose rather than obscure the trade-offs.

Despite this theoretical clarity, the empirical interaction effects of combining defense methods across all three axes remain poorly mapped. Most published work optimizes a single axis in isolation, and the few studies that combine methods (e.g. DP + robust aggregation) observe that their costs compound in non-obvious ways: **Both DP noise injection and Byzantine-robust aggregators clip or perturb gradients, leading to error amplification beyond what either mechanism incurs independently.** This project addresses that gap through systematic empirical experimentation.

2 Research Questions

The project is structured around three research questions

1. **Isolation:** What is the individual cost of each defense mechanism (DP, Byzantine-robust aggregation, gradient compression) in terms of accuracy, communication overhead, and attack success rate, and do our results reproduce the published baselines?

2. **Combination:** How do pairwise and full combinations of the three mechanisms interact? Do the costs compound additively, sub-additively, or super-additively?

3. **Mitigation:** Can simple adjustments — e.g. joint calibration of the DP clipping norm and the robust aggregation threshold — reduce the compounding effect without sacrificing formal guarantees?

3 State of the Art

3.1 The Trilemma: Theoretical Foundations

Allouah et al. (ICML 2023) [1] proves that any algorithm guaranteeing ϵ-local differential privacy and robustness against an η-fraction of adversarial machines must incur mean-estimation error $\Omega(\frac{\eta}{\epsilon^2} + \frac{d}{\epsilon^2 n} + \eta G^2)$. The cross-term $\frac{\eta}{\epsilon^2}$ shows: robustness and privacy interact multiplicatively, not independently. The paper also introduces the SMEA aggregator that matches this lower bound — at exponential computational cost — and the polynomial-time CAF aggregator as a practical approximation. A 2023 position paper by the same group [2] extends this framework to the efficiency axis and identifies the gap between theoretical optimality and empirical scalability as an open problem.

3.2 Privacy: Differential Privacy in FL

In 2016, Abadi at al introduced DP-SGD [3], which clips per-sample gradients to bound sensitivity and adds calibrated Gaussian noise. In the FL context, McMahan et al. [4] established FedAvg as the baseline aggregation strategy. Central DP applies noise at the server after aggregation; local DP applies noise at each client before upload. The Flower framework [5] natively supports both modes, and Opacus [6] provides the DP-SGD implementation with Rényi DP accounting [7].

3.3 Robustness: Byzantine-Tolerant Aggregation

Blanchard et al. [8] introduced **Krum**, the first provably Byzantine-tolerant aggregation rule (selecting the update closest to its neighbors). Cao et al. [9] proposed **FLTrust**, which bootstraps trust from a small server-held root dataset and performs better under data heterogeneity. On the attack side, Fang et al. [10] demonstrated that local model poisoning attacks can circumvent most existing defenses under weak attacker fractions. The BLADES benchmark [11] provides a unified, reproducible evaluation suite for both attacks and defenses.

3.4 Performance: Communication Efficiency and Model Performance

Gradient compression via Top-k sparsification [12], [13] reduces per-round communication cost by transmitting only the k largest gradient components. Recent work on selective low-rank adaptation (LoRA) combined with DP [14] demonstrates that the performance–privacy axis can be partially addressed through model reparametrization. Communication cost is directly tied to model dimensionality d, which appears in the theoretical lower bound of as the term$\frac{d}{\epsilon^2 n}$, confirming efficiency as a fundamental third axis.

3.5 Research Gap

The literature on combined mechanisms is sparse. Guerraoui et al.[15] asked 'Differential Privacy and Byzantine Resilience in SGD: Do They Add Up?' and found that naive combination degrades performance beyond either mechanism alone. No subsequent work has systematically mapped combinations of DP $\times$ robustness $\times$ compression on a common benchmark with joint reporting across all three axes which is the experimental design proposed here.

4 Methodology

4.1 Method Selection

To keep the experimental grid tractable while remaining representative, one method per trilemma axis is selected, chosen for theoretical grounding, open-source availability, and prevalence in the literature (Table 2).

Axis	Method	Key Parameters	Why do we choose this?
Baseline	FedAvg [4]	-	Commonly used for benchmarking
Privacy	DP-SGD [3]	$\epsilon \in \{1, 5, 10\}$, $\delta = 10^{-5}$	good standard, tunable privacy budget, Opacus integration with Flower
Robustness	FLTrust [9]	Server root dataset size	Stronger than Krum under heterogeneity; available in BLADES
Performance	Top-k sparsification [13]	$k \in \{1\%, 10\%\}$ of d	Simple, compatible with DP and robust agg.; directly controllable

Table 2: Method Selection

4.2 Experimental Setup

The experimental grid consists of all 2^3 on/off combinations of the three mechanisms run on two datasets:

- MNIST — simple baseline, reproducibility check against BLADES published numbers
- CIFAR-10 — tests scalability; used in the position paper's empirical evidence table [2]

Additional parameters (fixed across runs): Number of clients, Byzantine fraction, number of communication rounds, model architecture

4.3 Software Stack

The following open-source tools are integrated into a single experimental pipeline:

- Flower[1] (flwr) [5] — FL orchestration, client/server simulation, strategy interface
- Opacus[2] [6] — DP-SGD, Rényi DP accounting, gradient clipping
- BLADES[3] [11] — Byzantine attack and FLTrust defence implementations; PyTorch/Ray backend compatible with Flower
- Custom — Top-k gradient sparsification wrapper

4.4 Metrics and Reporting

Every configuration reports all three axes simultaneously:

- Privacy: (ϵ, δ)-DP budget consumed after T rounds; membership inference attack success rate (shadow model)
- Robustness: Attack success rate (ASR) under label-flipping; accuracy gap vs. clean FedAvg
- Performance: Test accuracy; bytes per client per round; wall-clock training time

Results can be visualised on a trilemma simplex / radar chart, positioning each of the 8 configurations relative to the three axes — directly analogous to Figure 1 of [2]:

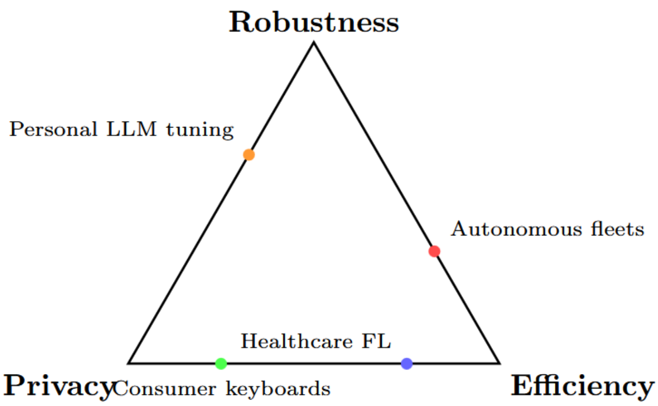

Figure 1: Trilemma Visualization from [2]

[1]https://flower.ai/
[2]https://opacus.ai/
[3]https://github.com/lishenghui/blades

5 Work Plan

1. Read into core literature
2. Set up software stack
3. Reproduce FedAvg and FLTrust baselines on MNIST
4. Run each mechanism independently (RQ1), sanity-check against published results
5. Run all 2^3 combinations, record accuracy, ASR, communication cost, privacy budget; identify compounding patterns, especially DP + FLTrust interaction (RQ2)
6. Implement joint clipping-norm calibration heuristic; test against worst-case combination (RQ3)

6 Expected contributions

- A systematic empirical mapping of all 8 combinations of DP $\times$ Byzantine robustness $\times$ gradient compression on MNIST and CIFAR-10, with joint reporting across all three trilemma axes — filling the gap identified by [2].
- Reproduction and extension of BLADES baseline results within the Flower ecosystem, demonstrating cross-framework compatibility.
- A falsifiable hypothesis and initial experimental test of joint clipping-norm calibration as a lightweight mitigation for compounding effects.
- A reproducible open-source experimental pipeline (Flower + Opacus + BLADES) for future work.

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